

Lecture 4 Part 1:

A Crash Course In

Experimental Design

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Remind of what we did last week

- We spoke about hypothesis and ethics
 - Why we need it, and examples of poor ethical behaviour
 - Importance of ethics within computer science
- We spoke about arguments (philosophically)
 - Inductive vs deductive inference
 - Informal and formal fallacies
 - False dilemmas
 - Straw man arguments
- And we also spoke about possibility vs probability

Contents

- Overview of the video content A crash course in experimental design
- Discussing experimental designs — by example

Video: Experimental Conditions & Variables

Variables and Factors in Experimental Design

Know what the following are:

- Experimental Objects
- Tasks / Processes
- Levels and Conditions
- Independent / Dependent Variables
- Confounding Factors
- Control Variables
- Random Variables

You need to know what these are, and be able to identify them in an experimental design

Variables and Factors in Experimental Design

Need to know these because:

- This is the language of experimental design
- A methodological way of reporting your experiment
- Enables precise critiques of experimental designs, e.g. “why isn’t X a confounding variable” is better than “seems like X might be important for some reason”

You need to learn the language of any topic you work in - this is some standard jargon across all of research regardless of topic

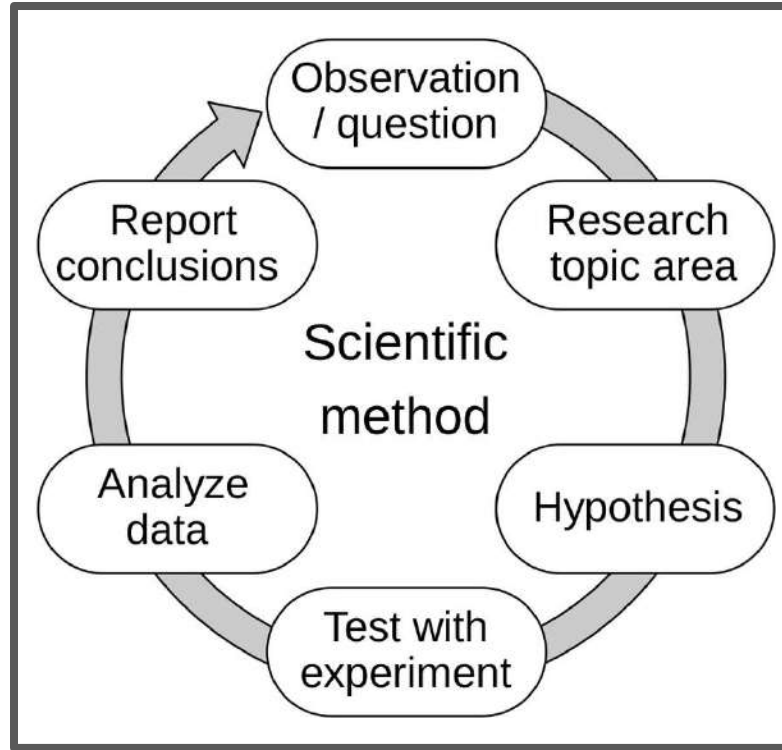
Video: Experiments In Computer Science

VIDEO — Experiments in CS

- Give you an overview of the many types of CS research (going on here at Glasgow)
- Introduce **between** and **within subjects** experimental design

The Scientific Method & Experimental Design

Previously we introduced the scientific method



And the creation of a “hypothesis”

Hypothesis: a **specific** and **testable** explanation for a phenomena.

- *Specific*: explains **what you expect to happen** in a certain circumstance;
- *Testable*: is **falsifiable** (can be disproven by empirical data);
- *Parsimonious*: is as **simple** as feasible (remember **Occam's Razor**);
- *Predictive*: makes a **prediction** that can be tested.

Today we will look at how to
experimentally investigate the
hypothesis

Experimental Design

Experiments

A key part of the scientific method is testing your hypotheses

- Experiments are used to gather empirical data/measurements

We need to use carefully controlled experiments

Experiment designs vary — all serve the same purpose: gathering data against which we can test our hypotheses

The experimental factors checklist

- Research Question / Hypothesis
- Independent / Dependent Variables
- Control Variables Random Variables
- Confounding Factors Participants
- Within/Between Subject Designs

Need to think about all of these aspects – we will cover these today

Independent Variables

Independent variable: what you deliberately change in the experiment

- A variable that the experimenter controls
- Have different “levels”
 - E.g. configuration 1, configuration 2, configuration 3, etc.
- If multiple independent variables — a “condition” is a combination of independent variable levels

Independent Variables

What do you mean by levels of independent variable?

- Each test condition

Example: If “Input Device” is the independent variable examples of levels might be “mouse” and “trackpad” and “eye-gaze”, etc.

Dependent Variables

Dependent variable: the data you are collecting

- A variable whose value(s) depend on the independent variable

Any observable, measurable behaviour can be a dependent variable

- Task execution time, typing speed, accuracy, error rate, questionnaire responses, comfort, emotional response, etc.

Independent and Dependent Variables

- When designing an experiment, an experimenter needs to:
 - Identify 1 or more independent variables
 - These are what the experimenter is going to manipulate
 - Identify 1 or more dependent variables
 - These are the data that are going to be measured / collected

What do we want an experiment to show?

A **change** in the values of the **independent variable** will result in a **change** in the values of the **dependent variable**

We want to be sure that:

- **Any change** in the value of the **dependent variable** is a result of the change in the value of the **independent variable**
(rather than a result of change in the value of other factors)

Control Variables

Control Variables: factors that do not change to ensure a “fair” comparison

- These are not under investigation (i.e. are not IVs) but they might influence participant behaviour (DVs)
- Experimenters control these variables to prevent their influence by setting up their study in a controlled environment

Random Variables

Random Variables: factors you knowingly allow to randomly change/vary

- Instead of attempting to control for everything, it is often better to allow some variables to vary randomly to generalise the results (and because controlling for everything is probably impossible)
 - E.g. Height, finger size, hand size, etc.

Each study requires judgement about the trade off between control and allowing random variation

Confounding Factors

Confounding Factors: factors that change and which impact the results (usually only identified after you have collected the data)

- Factors which could have influenced the relationship between the independent and dependent variable

Confounding Factors

In some cases, a circumstance or condition will change systematically with the independent variable

- E.g. practice, different types of measurement for levels of the IV, prior experience with an interface (keyboard example)

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Confounding Factors

Such variables are confounding because they prevent the possibility of a cause-and-effect relationship being inferred from the results

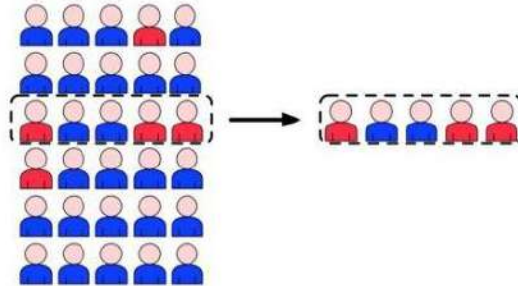
- So, they need to be controlled for
- This is a central skill in experimental design

Participants

In order to correctly assume that research results apply to people other than those recruited you must:

Recruit people from the population you want to investigate

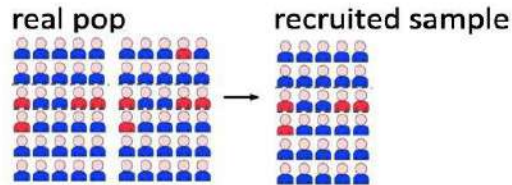
Recruit a sufficient number of participants



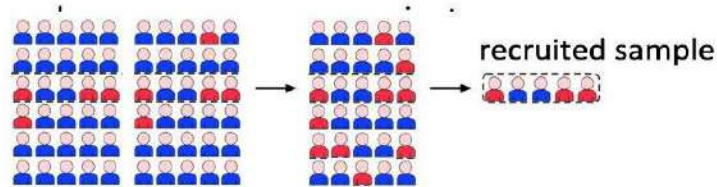
Participants

Recruitment methods

Ideal = participants drawn at random from a population



In practice = convenience sampling



How might these people differ from the population you are trying to generalise your findings to?

Participants

How many people should you recruit?

More is better

However there is a balance between representing the population and practical, and sometimes ethical considerations

Practical: Not much time to recruit (e.g. in student project), population difficult to access, more testing delays product going to market

Ethical: Study puts participants under burden; continuing study beyond necessary delays useful intervention or technique that can improve access.

A note about human characteristics

Since we are talking about user experiments

For user studies (experiments involving human participants) differences in human characteristics are factors in experimental design:

- Age, gender, height, weight, left/right-handed, personality traits, political viewpoints, ethnicity, cultural background, first language, etc.